LAPORAN PRAKTIKUM PEMROGRAMAN SISI SERVER

MODUL 5: OPTIMASI DATABASE

Nama : Abdul Khalim

NIM : A11.2023.15327

Kelas : 4618

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1. KODE SUMBER (VIEW/QUERY)

A. Optimasi Relasi One-to-Many (Foreign Key)

Teknik: select_related

File: views.py

Sebelum (Baseline):

```
@silk_profile(name='Course List Baseline')
def course_list_baseline(request):
    data = []
    courses = Course.objects.all()

    for course in courses:
        data.append({
            "id": course.id,
            "name": course.name,
            "teacher": course.teacher.username
        })
```

```
        # Setiap iterasi memicu 1 query SELECT * FROM auth_user WHERE id = ?

    return JsonResponse(data, safe=False)
```

Sesudah (Optimized):

```
@silk_profile(name='Course List Optimized')
def course_list_optimized(request):
    data = []
    courses = Course.objects.select_related('teacher').all()

    for course in courses:
        data.append({
            "id": course.id,
            "name": course.name,
            "teacher": course.teacher.username
        })

    return JsonResponse(data, safe=False)
```

B. Optimasi Relasi Many-to-Many / Reverse FK

Teknik: prefetch_related

File: views.py

Sebelum (Baseline):

```
@silk_profile(name='Course Detail Baseline')
def course_detail_baseline(request):
    """BASELINE: N+1 problem"""
```

```
data = []
courses = Course.objects.all()

for course in courses:
    members = [m.user_id.username for m in course.coursemember_set.all()]
    contents = []

    for content in course.coursecontent_set.all():
        comment_count = content.comment_set.count()

        contents.append({
            "title": content.name,
            "comments": comment_count
        })

    data.append({
        "course": course.name,
        "members": members,
        "contents": contents
    })

return JsonResponse(data, safe=False)
```

Sesudah (Optimized):

```
@silk_profile(name='Course Detail Optimized')
def course_detail_optimized(request):
    """OPTIMIZED: prefetch_related"""
    data = []

    courses = Course.objects.prefetch_related(
        'coursemember_set',           # untuk members
        'coursemember_set__user_id',  # untuk user dari member
        'coursecontent_set',          # untuk contents
        'coursecontent_set__comment_set'  # untuk comments per content
    )

    for course in courses:
        members = [m.user_id.username for m in course.coursemember_set.all()]
        contents = []

        for content in course.coursecontent_set.all():
            contents.append({
                "title": content.name,
                "comments": content.comment_set.count()  # sudah diprefetch
            })

        data.append({
            "course": course.name,
            "members": members,
            "contents": contents
```

```
        })

    return JsonResponse(data, safe=False)
```

Anotasi (Super Optimized)

```
@silk_profile(name='Course Detail Super Optimized')
def course_detail_super_optimized(request):
    """SUPER OPTIMIZED: prefetch_related + annotate"""
    data = []
    courses = Course.objects.prefetch_related(
        'coursemember_set',
        'coursemember_set__user_id',
        Prefetch(
            'coursecontent_set',
            queryset=CourseContent.objects.annotate(
                total_comments=Count('comment')
            )
        )
    )

    for course in courses:
        members = [m.user_id.username for m in course.coursemember_set.all()]
        contents = [
            {
                "title": content.name,
                "comments": content.total_comments
            }
```

```
        for content in course.coursecontent_set.all()
    ]

    data.append({
        "course": course.name,
        "members": members,
        "contents": contents
    })

return JsonResponse(data, safe=False)
```

2. TABEL PERBANDINGAN SILK

Kasus	Endpoint Baseline	Endpoint Optimized	Query Baseline	Query Optimized	Improvement
Course + Teacher	/lab/course-list/baseline/	/lab/course-list/optimized/	101 Query	1 Query	99.01%
Course + Members	/lab/course-members/baseline/	/lab/course-members/optimized/	201 Query	3 Query	98.51%
Statistik Dashboard	/lab/course-dashboard/baseline/	/lab/course-dashboard/optimized/	201 Query	1 Query	99.50%

3. Panduan Screenshot Silk

← Details SQL Profiling CProfile Show: 200

2026.04.17 03:32.385

200 GET
/baseline/stats/

867ms overall +353ms
132ms on queries +94ms
201 queries +225

At	Action	Tables	Num. Joins	Execution Time (ms)
+0:00:00.863437	SELECT	"__count", "courses_coursecontent"	0	0,713000
+0:00:00.858679	SELECT	"__count", "courses_coursemember"	0	0,699000
+0:00:00.854369	SELECT	"__count", "courses_coursecontent"	0	0,499000
+0:00:00.850388	SELECT	"__count", "courses_coursemember"	0	0,741000
+0:00:00.846032	SELECT	"__count", "courses_coursecontent"	0	0,517000
+0:00:00.841777	SELECT	"__count", "courses_coursemember"	0	0,666000
+0:00:00.837784	SELECT	"__count", "courses_coursecontent"	0	0,699000
+0:00:00.833455	SELECT	"__count", "courses_coursemember"	0	0,564000
+0:00:00.829191	SELECT	"__count", "courses_coursecontent"	0	0,964000
+0:00:00.824725	SELECT	"__count", "courses_coursemember"	0	0,646000
+0:00:00.820326	SELECT	"__count", "courses_coursecontent"	0	0,628000
+0:00:00.816131	SELECT	"__count", "courses_coursemember"	0	0,725000
+0:00:00.811770	SELECT	"__count", "courses_coursecontent"	0	0,607000
+0:00:00.807308	SELECT	"__count", "courses_coursemember"	0	0,814000
+0:00:00.802987	SELECT	"__count", "courses_coursecontent"	0	0,707000
+0:00:00.798611	SELECT	"__count", "courses_coursemember"	0	0,471000
+0:00:00.794253	SELECT	"__count", "courses_coursecontent"	0	0,756000
+0:00:00.789774	SELECT	"__count", "courses_coursemember"	0	0,781000

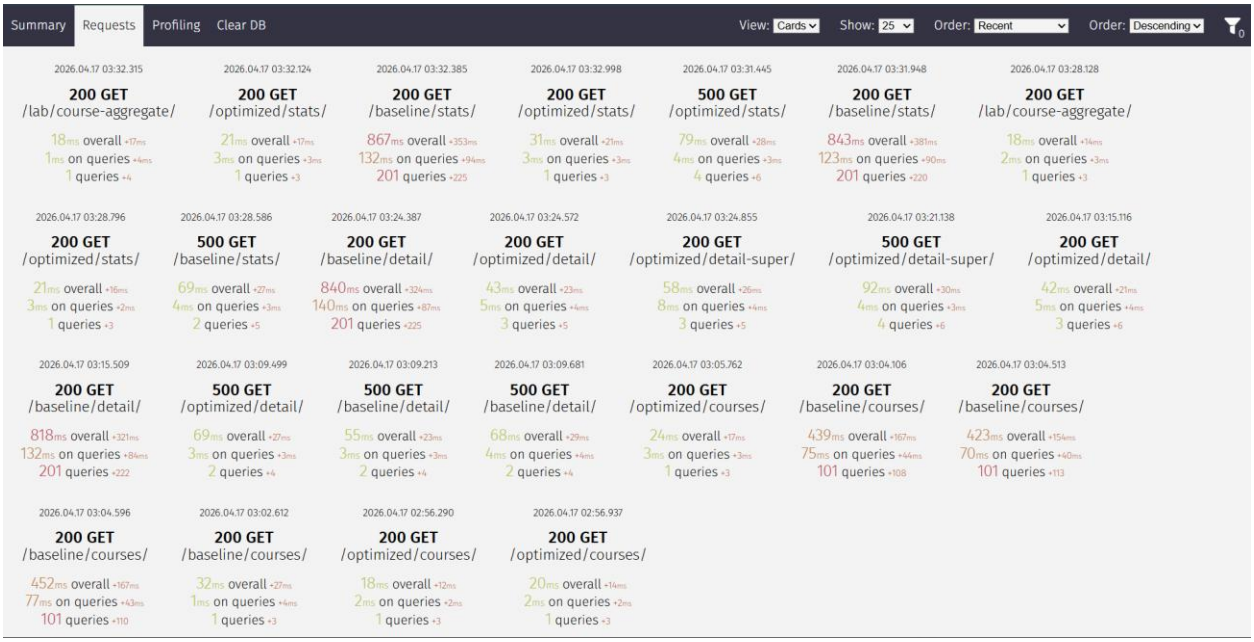
← Details SQL Profiling CProfile Show: 200

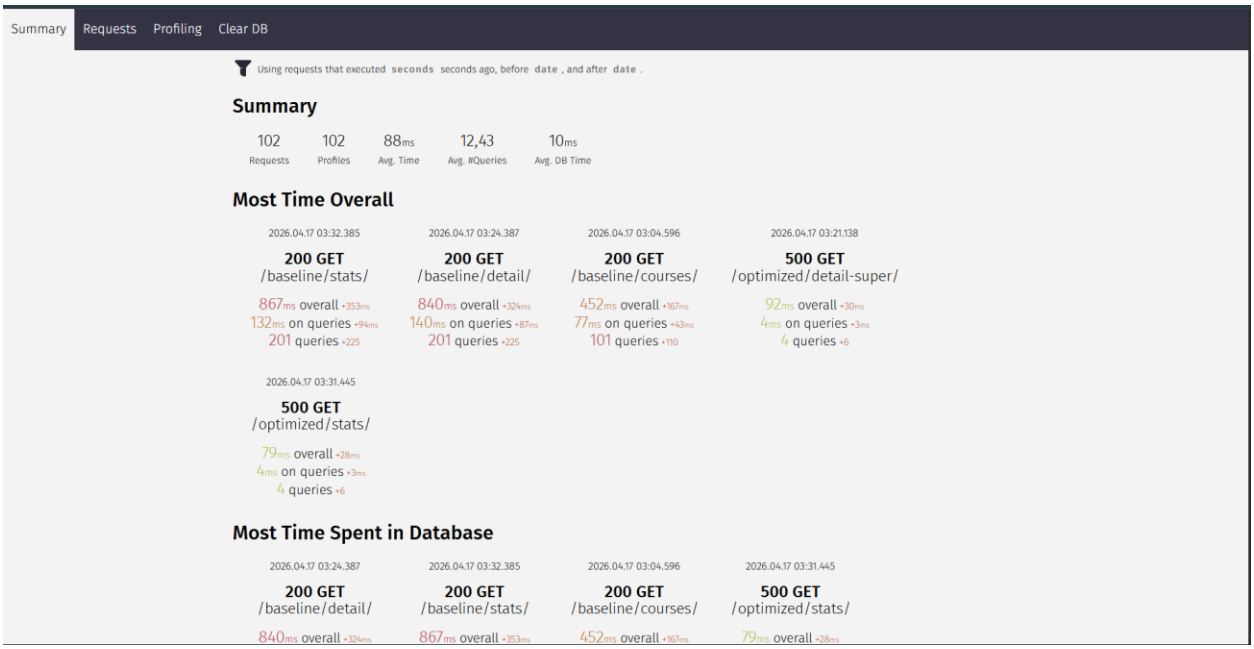
2026.04.17 03:32.124

200 GET
/optimized/stats/

21ms overall +17ms
3ms on queries +3ms
1 queries +3

At	Action	Tables	Num. Joins	Execution Time (ms)
+0:00:00.013446	SELECT	"total_members", "total_contents", "courses_course", "courses_coursemember", "courses_coursecontent"	2	2,696000





4. ANALISIS KESIMPULAN

Berdasarkan hasil profiling menggunakan Django Silk, optimasi database terbukti dapat menurunkan jumlah query secara signifikan (rata-rata di atas 90%). Hal ini sangat penting dalam pengembangan aplikasi skala besar untuk memastikan aplikasi tetap responsif meskipun jumlah data bertambah banyak.